Direct and centered Diophantine consistency equations

AI assisted research prepared by dot for Vladimir Reshetnikov

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Abstract

We give a fully materialized ordinary quartic polynomial whose lack of integer zeros is provably equivalent in ZFC to syntactic consistency of ZFC. The construction starts directly from the fixed numerical C222 proof calculus, uses a coefficient-height certificate for an unordered proof, and eliminates one bounded family by explicit CRT, factorial, binomial, power and Pell representations. It uses neither K215 machine simulation nor Jones's universal polynomial. The result has 27,916 unrestricted integer variables, 758,275 nonzero monomials, and largest absolute coefficient 320,307,372,009,918. A companion has 16,036 nonnegative variables. This is an architectural completion, not a variable-count record.

Separately, integral square completion in a final radical norm gives four 10-integer cost–degree tradeoffs, down to degree 1,080,888. A cofinal auxiliary Pell orbit yields an 11-positive-variable equation of degree 211 with 113 live arithmetic operations. Reversible norm-row reduction gives 21 unrestricted integers at degree 16. The complete literal equations, actual fixed integer payloads, mathematical proof interfaces, independent checkers and deterministic replay are supplied. None of these claims is a new end-to-end Lean or Rocq certificate or a proof of consistency.

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1 Results and the meaning of simplicity

The target statement is the usual syntactic sentence Con(ZFC): there is no finite ZFC proof of a fixed contradiction. Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$, whereas a positive variable ranges over $\mathbb{Z}_{>0}$. For every completed equation identified below, ordinary mathematics formalizable in ZFC proves

$$\mathrm{Con(ZFC)} \quad \longleftrightarrow \quad \neg\exists \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) = 0, \qquad D \in \{\mathbb{N}, \mathbb{Z}, \mathbb{Z}_{>0}\}.$$

This biconditional does not establish either side. A fixed coefficient may be enormous without being an unknown; a circuit register may abbreviate a polynomial without being a quantified variable. The report keeps these notions separate.

There is no single total order called "simplest." The following completed round-twelve points optimize different resources. A binary arithmetic operation means one addition, subtraction or multiplication, including a square or a scalar product; copying and reading a fixed integer leaf cost zero.

Construction	Domain	Variables	Degree	Live operations
Direct proof source	$\mathbb{Z}$	27,916	4	expanded polynomial
Direct proof source	$\mathbb{N}$	16,036	4	quadratic residual list
Centered auxiliary row	$\mathbb{Z}$	21	16	not optimized
Two small Pell gaps	$\mathbb{Z}_{>0}$	11	211	113
Short guard and polarization	$\mathbb{Z}$	10	1,245,324	471
First radical center	$\mathbb{Z}$	10	1,179,788	485
Second radical center	$\mathbb{Z}$	10	1,114,252	496
Further integral correction	$\mathbb{Z}$	10	1,080,888	545

For a compact evaluable ten-integer equation, the 471-operation point is the lowest-cost delivered choice. If degree matters more, use one of the centered alternatives. For a transparently completed direct route independent of the machine/Jones architecture, use the large quartic. Neither recommendation asserts a global optimum.

1.1 What is new in this iteration

The direct route was previously a proposal ending at an unmaterialized compiler interface. Its final ordinary equation is now present, with every coefficient and monomial, and each primitive interface independently audited. The common witness bound, fixed binary height bound, affine-wire elimination, and restriction of nonnegativity certificates to semantic inputs reduce the actual compiled size. The mathematical ingredients of coefficient-height encoding, CRT and Pell representation themselves are inherited.

In the small-equation branch, the new work is the source-aware short guard, three explicit integral square completions, simultaneous transport of two small Pell gaps with a cofinality proof, and the compatible auxiliary-norm row operation for 21 integers at degree 16. A 12-positive-variable degree-70 100-operation circuit is also supplied, but it materializes a prior manuscript formula and is not claimed as a new theorem.

1.2 Comparison boundaries

The repository comparison is pinned to ProveIt commit 9a33342a03d19cf5b061b2083c6b5d3f27f752ed. Six authenticated incoming archives were also inspected as inert data. A later Library supplement, *Pell Congruences*, is separate evidence, not content attributed to that repository pin. It already

contains $(n, d) = (20, 20), (19, 28), (18, 32), (17, 88)$ over unrestricted integers. Consequently our $(20, 20)$ charts are alternatives, and our $(19, 32)$ and $(18, 88)$ charts are dominated. Only the new $(21, 16)$ point is promoted in that low-degree comparison. The supplement's 9-positive-variable 1,158,228-degree 726-operation claim is distinct from the earlier 1,753-operation packet; it is not treated here as a new independently reconstructed result. The repository's 9-positive-variable 440-operation point has degree 1,845,940. Fewer variables, smaller degree, smaller circuits and smaller numeral payloads remain distinct objectives.

The source audit records exact archive and retained-file hashes. Hashes establish identity and provenance, not mathematical correctness. No upstream program was executed during this research replay.

2 A fully materialized equation from unordered proofs

This section closes a different gap from the low-variable record search. The height-certificate source previously ended at an unelaborated compiler interface. We now give an ordinary polynomial with every coefficient and monomial materialized. It is independent of the K215 transition table and of Jones's universal polynomial. Its size is large: the purpose is a complete, reproducible direct baseline, not a claim of a new variable record.

The recommended artifact is the literal ordinary polynomial Q in `direct/fixed_base/equation_integer.json`. It has 27,916 unrestricted integer variables, exact degree four, 758,275 nonzero monomials, and maximum absolute coefficient 320,307,372,009,918. In ZFC,

$$\mathrm{Con}(\mathrm{ZFC}) \quad \longleftrightarrow \quad \neg\exists \boldsymbol{x} \in \mathbb{Z}^{27916}\, Q(\boldsymbol{x}) = 0.$$

The natural-domain companion has 16,036 nonnegative variables. It is specified exactly as the sum of the squared residuals in `fixed_base/residuals_affine.json`; rebuilding also emits its collected sparse polynomial `equation_affine.json`. Here $\mathbb{N} = \{0, 1, 2, \ldots\}$. Each sparse entry `[coefficient,[indices]]` denotes that coefficient times the product of the listed variables; repeated indices denote repeated factors and indices start at zero. All coefficients are literal integers, not bounded search recipes. The compact archive retains the full integer polynomial and exact residual/circuit data; rebuilding restores redundant expanded intermediates and checks their recorded SHA256 hashes.

The derivation is mathematical; we do not claim a newly kernel-checked end-to-end translation. We first explain the two-stream source and then replace its height stream by a fixed binary bound. The final measurements above include this reduction and affine-wire elimination.

2.1 The fixed proof source

The inherited file `fixed_base/horn_shared_source.json` specifies twelve padded single-conclusion rules. Their local relation is

$$H(p, q, c; \boldsymbol{w}) = \prod_{j=0}^{11} \left(\sum_{r \in \mathcal{R}_j} r(p, q, c, \boldsymbol{w})^2 \right) = 0.$$

There are seventeen shared nonnegative witnesses. The residual lists $\mathcal{R}_j$, including the fixed bundled axiom, are included literally. Their expanded sums of squares are independently reproduced exactly by the checker. The historical JSON field name `quadratic_residuals` is not a degree assertion: the bundled branch also contains quartic residuals. The total product has degree fifty. The compiled

endpoint deliberately uses this twenty-coordinate local source (three labels and seventeen Horn witnesses), not the earlier twenty-three-coordinate degree-fourteen selector variant. After adding six beta coordinates and three label-bound slacks, the bounded row has twenty-nine witnesses. Reducing local polynomial degree need not reduce the final quartic: each extra bounded witness adds arithmetic certificates, whereas circuit flattening already makes the global degree four. The first rule is truth $(0,0,0)$; zero premises pad axioms and one-premise rules. Target code one is falsity.

For clarity, the calibration does not pass through execution of K215. The repository theorem `c222_iff_zf_refutation` in `KEndpoint.lean`, followed by `zf_refutation_iff_zfc_refutation` in `ConstructibleChoice.lean`, identifies derivability of target one in this numerical calculus with a syntactic ZFC refutation. The exact numeric constructors and twelve rules are in `KCode222.lean` and `Code254.lean`. These are dependencies of the mathematical calibration, not claims that the new Python compiler has been checked by Lean.

A finite family of valid rules has a *flow certificate* for one when, for every positive label j,

$$\#\{i : c_i = j\} = [j = 1] + \#\{i : p_i = j\} + \#\{i : q_i = j\}. \tag{F}$$

Derivability implies (F) by unfolding a proof tree. Conversely, if one is not derivable, sum these balances over the nonderivable labels in the finite support. Each valid single-conclusion rule contributes at most zero: a nonderivable conclusion has a nonderivable premise. The right side requires a positive net contribution at one, a contradiction. This argument permits cycles and repeated premises but uses both nonnegative multiplicities and the single-conclusion hypothesis.

2.2 The coefficient-height certificate

Given N rule instances, put

$$A(X) = \prod_{i<N} (1+c_iX), \quad B(X) = (1+X) \prod_{i<N} (1+p_iX)(1+q_iX), \quad R_N = 2 \prod_{i<N} ((1+p_i)(1+q_i)+c_i).$$

Every coefficient of $A - B$ has absolute value at most R_N. Consequently, for $M = R_N + 1$, equality $A(M) = B(M)$ forces $A = B$: the least nonzero coefficient of a nonzero polynomial vanishing at M would be a nonzero multiple of M of absolute value smaller than M. The distinct roots $-1/j$ recover (F) from the polynomial identity.

This yields the two-accumulator source

$$\begin{aligned} &S_0 = M + 1, \quad S_N = 1, \quad R_0 = 2, \quad M = R_N + 1, \\ &S_{i+1}(1 + c_iM) = S_i(1 + p_iM)(1 + q_iM), \\ &R_{i+1} = R_i((1 + p_i)(1 + q_i) + c_i), \qquad H(p_i, q_i, c_i; \boldsymbol{w}_i) = 0. \end{aligned}$$

All factors are positive. Telescoping gives $A(M) = B(M)$; the second accumulator supplies the coefficient bound. Conversely, expand a proof tree in open-obligation order, compute the complete R_N first, choose $M = R_N + 1$, and let S_i be the product of the open-obligation factors. There is no circular choice of the coefficient bound.

Encode the two sequences using a common beta scale:

$$\beta(x, y, i) = x \bmod (1 + (i + 1)y).$$

Every pair of finite natural sequences admits such an encoding with separate x, z and one common y, by choosing y large and divisible by $N!$, then using the Chinese remainder theorem. The factorial

divisibility here is used only in the existence proof; it is not an additional universally quantified condition in the final source.

The five external parameters are N, x, z, y, M_0, with $M = M_0 + 1$. Four beta endpoint equations are imposed. In the one bounded row block, the four beta values are replaced by four triples (r, q, s) satisfying

$$x = qd + r, \qquad d = r + s + 1, \qquad d = 1 + (i + 1)y,$$

with the appropriate code and index. Thus the row has exactly $20 + 4 \cdot 3 = 32$ natural witnesses: twenty for labels and Horn witnesses, and twelve for the beta graphs. There is a fixed signed polynomial $P = e - f$ with e, f having nonnegative coefficients such that the source is

$$\text{four endpoint equations} \quad \wedge \quad \forall i < N \; \exists \boldsymbol{w} \in \mathbb{N}^{32} \quad e(i, \boldsymbol{w}) = f(i, \boldsymbol{w}). \tag{S}$$

For this two-stream comparison, e, f can be represented as nonnegative arithmetic DAGs. The delivered `fixed_base/row_source.json` instead contains the twenty-nine-witness specialization described below. Both use the following identities without expanding the large Horn product: $(e - f) + (g - h) = (e + g) - (f + h)$ and $(e - f)(g - h) = (eg + fh) - (eh + fg)$.

2.3 One uniform witness bound and one CRT block

This materialization uses a small improvement to the generic bounded universal compiler: one common bound U replaces thirty-two independent coordinate bounds, and the factorial $(U + 1)!$ is computed only once. Set

$$T = (e(N, U, \ldots, U) + f(N, U, \ldots, U) + N)!, \qquad D_j = 1 + (j + 1)T, \qquad B = \prod_{j<N} D_j.$$

The bounded block in (S) is equivalent to the following finite existential conditions, with encoded coordinates $a_1, \ldots, a_{32}$ and one encoded index v:

$$\begin{aligned} B \mid (U + 1)! \binom{a_i}{U + 1} \qquad & (1 \le i \le 32), \\ B \mid 1 + (v + 1)T, \qquad & \\ e(v, \boldsymbol{a}) \equiv f(v, \boldsymbol{a}) \pmod{B}. \qquad & \end{aligned}$$

For soundness, fix $j < N$ and take a prime $p \mid D_j$. Since $\gcd(p, T) = 1$ and T is the displayed factorial, p exceeds its factorial argument. Divisibility of each falling factorial supplies a residue $w_i \le U$ with $a_i \equiv w_i \pmod{p}$. This statement also handles truncated natural subtraction: if a vanishing factor arises from $a_i < t$, choose $w_i = a_i$. The index congruence gives $v \equiv j \pmod{p}$. Hence $e(j, \boldsymbol{w}) \equiv f(j, \boldsymbol{w}) \pmod{p}$. Monotonicity of nonnegative-coefficient polynomials bounds both values strictly below p, so they are equal. For completeness choose U above all finitely many row witnesses. The D_j are pairwise coprime because T is divisible by all positive differences below N; CRT supplies the encoded columns and index. The argument is also valid for $N = 0$. The original source's endpoints themselves exclude $N = 0$.

This is a uniform-bound specialization of the pinned theorem `boundedForallCertificate`. It removes no semantic requirement and needs no unproved new representation theorem.

2.4 Eliminating every nonpolynomial operation

The file `fixed_base/build_direct.py` implements the following explicit identities. Remainders and quotients always have positive denominators and are encoded by $n = qd + r$, $d = r + s + 1$. Congruence modulo d, including $d = 0$, is encoded by $a + du = b + dv$ with two natural quotients. If $Q = ((N+1)T + 1)^N + 2$, then

$$B = (T^N N! \binom{Q+N}{N}) \bmod (TQ - 1).$$

The denominator is positive since $T \geq 1$ and $Q \geq 3$. The binomial and factorial identities are

$$\binom{n}{k} = \left\lfloor \frac{(2^n+2)^n}{(2^n+1)^k} \right\rfloor \bmod (2^n+1),$$
$$r = (n+2)^{n+2}, \quad C = \binom{r+n}{n}, \qquad n! = \left\lfloor \frac{(r+1)^n + C - 1}{C} \right\rfloor.$$

They include all boundary cases, including $k > n$ and $n = 0$. The predecessor $C - 1$ is represented by $C = C_- + 1$; here $C > 0$. The pinned `FactorialRepresentation.lean` proves these identities, and `BoundedForallRepresentation.lean` supplies the product formula.

Natural exponentiation is replaced by its fully explicit finite Pell certificate, not by an oracle. Write

$$\mathcal{N}(a, x, y) : \quad x^2 + y^2 - a^2 y^2 = 1.$$

The graph of $(x, y) = (X_n(a), Y_n(a))$, for $a > 1$, is given by $n \leq y$ and the disjunction of $x = 1, y = 0$ with the existence of u, w, s, t, b satisfying

$$\mathcal{N}(a, x, y), \quad \mathcal{N}(a, u, w), \quad \mathcal{N}(b, s, t), \quad b > 1, \quad w > 0,$$
$$b \equiv 1 \pmod{4y}, \quad b \equiv a \pmod{u}, \quad y^2 \mid w, \quad s \equiv x \pmod{u}, \quad t \equiv n \pmod{4y}.$$

This is exactly the pinned finite Matiyasevich certificate in `PellRepresentation.lean`. To represent $m = n^k$, use the disjunction

$$(k = 0 \wedge m = 1) \ \vee \ (k > 0 \wedge [(n = 0 \wedge m = 0) \ \vee \ (n > 0 \wedge \mathcal{C}(m, n, k))]),$$

where $\mathcal{C}$ asserts existence of w, a, t, z, x, y with the displayed Pell-coordinate graph at (a, k, x, y) and

$$x \equiv y(a \dot{-} n) + m \pmod{t}, \qquad 2an = t + n^2 + 1, \qquad m < t,$$
$$n \leq w, \qquad k \leq w, \qquad \mathcal{N}(w+1, a, wz).$$

Here $\dot{-}$ is truncated natural subtraction. Its graph is encoded by $(a = d + n)$ or $(a \leq n \wedge d = 0)$. These formulas are the exact `PowerRepresentation.lean` criterion. Strict and weak inequalities use natural slack variables; divisibility uses a natural quotient. Conjunctions are sums of nonnegative residual polynomials, and disjunctions are their products. Consequently inactive branch witnesses are harmless. In particular, no positivity assumption about an arbitrary power base is hidden in the generated polynomial.

The delivered fixed-base call ledger contains 108 power certificates, 108 indexed Pell certificates, 33 binomial calls, three factorial calls, and 72 division/remainder calls. The ledger records actual input and output node indices and existential ranges for each call.

2.5 Why the final polynomial is quartic

All signed expressions are maintained as differences of natural arithmetic circuits. Every addition or multiplication node receives a fresh natural wire and a defining residual of degree at most two. The final equality residuals are linear in the appropriate output wires. The delivered polynomial is the sum of the squares of all these residuals. A zero therefore enforces every circuit gate and every arithmetic certificate. Conversely, any certificate uniquely supplies all gate wires. Thus flattening preserves existence of zeros in both directions. A nonzero squared multiplication residual ensures exact degree four; the checker also verifies the degree of the fully collected polynomial.

Rebuilding emits `residuals.json` and `equation.json`, containing, respectively, the complete residual list and its fully collected sum of squares. Independent integer arithmetic reconstructs every gate and every coefficient. The test suite also expands all twelve Horn branches exactly, checks row evaluations against a separately written source expression, checks all zero-base and zero-exponent boundaries, and includes a genuine nonzero-exponent Pell witness for $1^1 = 1$. These finite checks support artifact fidelity; the all-input equivalence rests on the proofs above.

2.6 The fixed binary height bound

The completed compiler makes a further source improvement measurable. Choose an independent maximum label m, and impose $p_i, q_i, c_i \le m$ by three natural slack variables per row. Since $m + 1 \le 2^m$ for every natural m,

$$A(1) \le (m+1)^N \le 2^{mN}, \qquad B(1) \le 2(m+1)^{2N} \le 2^{2mN+1}.$$

It is therefore enough to set

$$M_0 = 2^{2mN+1}, \qquad M = M_0 + 1. \tag{BH}$$

The entire R stream and its beta code disappear. There is now only the S stream, with endpoints $\beta(x, y, 0) = M_0 + 2$ and $\beta(x, y, N) = 1$. Its local transition is unchanged. The row has $20+6+3 = 29$ witnesses instead of thirty-two: twenty for Horn/labels, six for two beta triples, and three label upper-bound slacks. One global power certificate imposes (BH). Thus eight power certificates disappear net, reducing 116 power calls to 108.

Completeness chooses a finite proof tree, then N and a bound m on its labels, then M_0 by (BH), then the S sequence, and finally its beta encoding and the uniform CRT witness bound U. In particular, m is *not* identified with U: the latter also bounds local remainders and quotients depending on M. Such an identification would introduce a circular, potentially impossible bound. Soundness uses the same coefficient-height lemma and flow theorem as before. For $N = 0$, (BH) gives $M_0 = 2$ and the two endpoints are contradictory.

The folder `direct/fixed_base/` gives this source and its complete polynomials. A second, slightly tighter but three-wire-larger version uses $M_0 = 2(m+1)^{2N}$; it is an intermediate comparison, not needed in the recommended release and not a distinct arithmetic theorem.

2.7 Eliminating affine wires and allowing unrestricted integers

Addition gates do not need existential output wires. Substitute their affine expressions into the next multiplication residual. Scalar products are affine too and are substituted likewise. Retain a wire only for a multiplication whose expanded expression is genuinely quadratic in previous retained wires and semantic inputs. Every residual still has degree at most two. The substitution is triangular and uniquely restores every omitted wire. All substituted affine forms have nonnegative coefficients, so natural witnesses are preserved.

Applied to the fixed-binary-height source, this gives the recommended natural polynomial (expanded by rebuilding as `fixed_base/equation_affine.json`):

Natural variables	16,036
Exact degree	4
Monomials	700,865
Semantic inputs	3,960
Retained nonlinear wires	12,076
Squared residuals	12,366
Maximum absolute coefficient	320,307,372,009,917

The number of monomials increases: fewer variables is not fewer expanded terms. The full affine substitution map is rebuilt and checked.

For a polynomial over unrestricted integers, only the 3,960 semantic inputs need explicit nonnegativity constraints. For each such input x, append

$$x - a^2 - b^2 - c(c+1) = 0.$$

Every natural x has this form: apply the three-square theorem to $4x+1$, whose representation has one odd and two even coordinates, then divide the identity by four after removing the odd square's constant. Conversely $a^2 + b^2 + c(c+1) \geq 0$ for integer a, b, c. Internal circuit wires may remain unrestricted because their triangular multiplication definitions force the unique nonnegative values after semantic inputs are nonnegative. Adding the squares of these 3,960 quadratic residuals gives the literal integer-domain polynomial in `fixed_base/equation_integer.json`:

27,916 unrestricted integer variables, exact degree four, 758,275 monomials.

Its maximum absolute coefficient is 320,307,372,009,918. Both lack-of-solutions statements are provably equivalent in ZFC to Con(ZFC). Neither count competes with the low-variable branch; the advance is an explicit, independently auditable ordinary equation from the direct proof source.

2.8 Scope and limitations

The direct equation is much larger than the low-variable Pell-based constructions. It should not be described as a numerical record. Its contribution is completion: no bounded quantifier, variable exponent, factorial, binomial coefficient, unknown compiler output, or unspecified coefficient remains in the final polynomial file. The fixed numeric proof source, the general arithmetic representations, and the flattening theorem are separate, auditable layers. Future source optimizations can now be measured against an actual ordinary equation rather than an interface.

3 The complete ten integer base formula

The following definitions make the norm transformations interpretable without confusing a circuit register with an extra variable. The ten unrestricted unknowns are

$$t, g_0, c_0, k_0, l, w, x, y, z, m.$$

The fixed H1868 encoding uses $\kappa = 2\,134\,940$ and the 31 weights w_i, 22 targets q_j, and three literal integer polynomials D_2, D_1, D_0 in `norm/data/code.json` and `norm/data/kernels.json`. Their

actual coefficient bytes, including all fixed source ports, are included and authenticated. Set

$$b = 6t + 4, \qquad g = bg_0, \qquad C = 6c_0 - 1, \qquad k = b(3k_0 + 2),$$
$$B_c = 2^{\kappa} b^2, \qquad \ell_c = \sum_i B_c^{w_i}, \qquad v_c = \ell_c + g.$$

$$c_c = D_2(B_c)v_c^2 + D_1(B_c)v_c + D_0(B_c), \qquad t_c = \sum_j B_c^{q_j},$$
$$N_0 = B_c^{117}, \qquad N_c = B_c^{1986},$$
$$\mu_c = N_0 - 1 - (b-1)\ell_c, \qquad S_c = g + N_0(2c_c + t_c),$$
$$T_c = \mu_c + N_0(B_c - 2)t_c - 2, \qquad X = N_c(S_c + T_c + 1) - S_c - 1,$$
$$Y = 2^{34\,159\,017} b B_c^{2109}, \qquad \mathcal{C} = bB_c^{116}.$$

These powers have fixed integer exponents. The data files are complete fixed coefficient lists, not unspecified parameters of an existence theorem. Continue with the integer polynomial registers

$$L = lY, \qquad U = 2LX, \qquad V = 4gwY, \qquad W = bw,$$
$$A = U(V + 1), \qquad B_1 = 2X + 1, \qquad S = 2A - 5.$$

$$T = 3WC - 2(W^2 - 1), \qquad P_h = (LX)UV,$$
$$K = X + 1 + k(2P_h - 2), \qquad Q = (P_h^2 - 1)K^2 + 1,$$
$$\Delta_g = g(\mathcal{C} - g)K^2 - \mathcal{C}^2(C - KL)^2, \qquad \xi = zx.$$

The new guard is justified in the next section. With $\mathrm{halfA} = (LX)(V + 1)$, so $A = 2\,\mathrm{halfA}$, put

$$D = (A^2 - 4)C^2 + 4, \qquad E_0 = C^2\xi, \qquad E = DE_0,$$
$$F = 4(A^2 - 4)C^4 D\xi^2 + 1, \qquad G = (\mathrm{halfA} - 1)(F - 1) - 1,$$
$$H = C + F(B_1 - C + 16Ey), \qquad I = (G^2 - 1)H^2 + 1,$$
$$\rho = T + Sy, \qquad \Sigma = \xi S^5, \qquad \Theta = (\rho^3 - \rho S^2 - S^3)(\rho^2 + 23S^2),$$
$$r = bQ, \qquad a_0 = r + (9\Delta_g - 3)(rz)^2, \qquad a_1 = DFI,$$
$$Z = \Sigma m - \Theta.$$

The base ordinary equation is

$$\left(Z^2 - \Sigma^2(a_0 + a_1)\right)^2 - 4\Sigma^4 a_0 a_1 = 0. \tag{N}$$

Every displayed symbol other than the ten unknowns is a fixed polynomial abbreviation. The exact optimized circuit, rather than this unoptimized rendering of its formula, determines the operation count.

3.1 Inherited semantic theorem and its use

The retained baseline proof establishes the preceding source decoder, the binomial bridge, signed-domain extraction and norm-saturated Lucas certificate. Its key reliability statement is that, under the sign and size conclusions forced by the guard, square DFI recovers the required Lucas index. The proof uses pairwise coprimality of D, F, I, Pell classification, index separation and rank divisibility.

Its completeness statement supplies ξ divisible by any prescribed positive integer and permits every residue class of y modulo ξ while keeping DFI square. This is the full residue interpolation property, not merely existence of one suitable y.

For arbitrary signed inputs, the two-root norm first gives square a_0, a_1 and $\Sigma \mid \Theta$. Coprime square-factor extraction and residues modulo three establish that b, Q are positive squares before guard signs are used. The guard then supplies the approximation and label bounds needed by the Lucas theorem and source decoder. Conversely canonical source witnesses, a Pell solution for a_0, the stated interpolation theorem, and the intersective quintic $(v^3 - v - 1)(v^2 + 23)$ produce an integer m solving (N). The full proof and earlier coefficient/compiler calibration are retained in `norm/BASELINE_PROOF.md` and `inherited/round11/`; they are mathematical dependencies, not assertions that imported formal files were rebuilt. The new guard is substituted only after proving its signed soundness and canonical completeness, as follows.

4 A source-aware approximation guard

Let $W = 116$, $B = 2^{2134940}b^2$, and $\mathcal{C} = bB^W$. The canonical digit code is

$$g = \sum_{i=1}^{31}(p_i - 1)B^{w_i}, \qquad 1 \le p_i \le b,$$

where the weights are distinct positive integers at most W. The coordinate a has weight W. Source equation 4 is $a = E+(N+1)s$; all three coordinates on the right are positive, so $a \ge 3$. Consequently $g \ge 2B^W$, and in particular $g \ge B^W$. The usual digit bound gives $g < \mathcal{C}$: indeed

$$g \le (b-1)\sum_{j=1}^{W} B^j < (b-1)\frac{B^{W+1}}{B-1} < bB^W,$$

since $B > b$. Moreover $B^W \ge b^2$, whence

$$g^3 \ge B^{3W} \ge b^2B^{2W} = \mathcal{C}^2.$$

Write $e = C - KL$ and replace the former approximation guard by

$$\boxed{R = g(\mathcal{C} - g)K^2 - \mathcal{C}^2e^2.}$$

For arbitrary integers with $\mathcal{C} > 0$, the condition $R > 0$ forces $K \neq 0$ and $0 < g < \mathcal{C}$. The identity

$$4g(\mathcal{C} - g) = \mathcal{C}^2 - (\mathcal{C} - 2g)^2 \le \mathcal{C}^2$$

then gives $4e^2 < K^2$. Thus the new guard implies the previous range and approximation conclusions, and in fact halves the permitted range.

Conversely take a canonical source code and the existing approximation $|e| < |K|/(4g)$, with $K \neq 0$. Since $\mathcal{C} - g$ is a positive integer,

$$\frac{R}{K^2} > g(\mathcal{C} - g) - \frac{\mathcal{C}^2}{16g^2} \ge g - \frac{g}{16} = \frac{15g}{16} > 0.$$

This proves completeness without a stronger approximation theorem or a new coefficient bound. The argument depends only on the verified top digit and existing radix and source conventions. Source encoding g is not the separate source coordinate also named g (whose weight is 9).

5 Centering the final radical

Here B abbreviates the already defined B_1. The inherited baseline equation uses ten unrestricted integer variables and has degree 1 245 324 and 480 binary arithmetic operations. The new short guard, polarization of the final norm, exact scalar retiming, and reuse of $uv = (LX)V$ in halfA $= LX + uv$ and $P_h = Uuv$ give a 471-operation equation at the same degree. A further idea trades a small increase in circuit size for a lower degree: translate the final witness by an integer polynomial approximating the largest radical.

Literal circuit	Operations	Exact degree
`integer10_polar.json`	471	1 245 324
`integer10_center0.json`	485	1 179 788
`integer10_center1.json`	496	1 114 252
`integer10_center2.json`	545	1 080 888

All four use ten unrestricted integers and the same actual fixed integer leaves. These are cost–degree tradeoffs, not uniform improvements or claims of minimality.

5.1 A universal coordinate-change lemma

Write $Z = \Sigma m - \Theta$, and let the previous final norm be

$$N = (Z^2 - \Sigma^2(a_0 + a_1))^2 - 4\Sigma^4 a_0 a_1.$$

Suppose integer polynomials P, J, independent of m, satisfy $a_1 = P^2 + J$. Define

$$T = Z + \Sigma P, \qquad U = Z^2 + 2\Sigma PZ + \Sigma^2(a_0 - J), \qquad N_c = U^2 - 4\Sigma^2 a_0 T^2.$$

Then $N_c(m) = N(m + P)$ identically. Indeed, $U = T^2 + \Sigma^2(a_0 - a_1)$ and polarization gives the identity. The substitutions $m \mapsto m \pm P$ are inverse bijections on integers, including when $\Sigma = 0$. Thus this change requires no new number theory, nonzero assumption, or loss of signed solutions. The cancellation in J is displayed explicitly instead of hidden in $a_1 - P^2$.

5.2 Two economical centers

Keep the inherited registers

$$D = (A^2 - 4)C^2 + 4, \quad E_0 = C^2\xi, \quad E = DE_0, \quad F = 4(A^2 - 4)C^4 D\xi^2 + 1,$$

$$G = (A/2 - 1)(F - 1) - 1, \quad H = C + F(B - C + 16Ey), \quad I = (G^2 - 1)H^2 + 1, \quad a_1 = DFI.$$

Here $A/2$ denotes the already available integer polynomial halfA, not a division operation in the circuit. The first center takes

$$R_0 = 2AE, \qquad J_{DF} = D - 16E^2.$$

The second, stronger center takes

$$R_0 = 2AE_0(D - 2C^2), \qquad J_{DF} = D + 16E_0^2((3C^2 - 4)D - 4C^2(C^2 - 1)).$$

In either case $DF = R_0^2 + J_{DF}$, hence

$$P = R_0 GH, \qquad J = J_{DF} G^2 H^2 - DF(H^2 - 1)$$

satisfy $a_1 = P^2 + J$. In the stronger version, DF has degree six in A, but its square defect has degree only two in A.

5.3 A further correction with integral coefficients

For the second center put $c_2 = C^2$, $h = A/2$, and define

$$\begin{aligned} L_* &= 1 + 16E_0^2(3c_2 - 4), & F_0 &= 1 + 64E_0^2(c_2 - 1),\\ \Omega_0 &= -64E_0(c_2 - 1), & H_0 &= C + F_0(B - C + \Omega_0 y),\\ V_0 &= -F_0H_0, & J_f &= E_0(c_2A^2 + 4 - 8c_2),\\ W_* &= B - C + 16Ey. \end{aligned}$$

Use the following division-free definition:

$$Q_* = (2h^2 - 2)EH - A\big[J_f(C + (F + F_0)W_*) + 4c_2F_0^2y\big].$$

One verifies $4c_2\xi AQ_* = GH - V_0$. This is an identity, not a quotient used to define Q_*. Its coefficients are integral in h, but need not be integral in an unconstrained formal variable A. Finally put

$$\begin{aligned} P &= R_0GH + L_*Q_*,\\ J_0 &= -4(c_2 - 1) - 256E_0^2(c_2 - 1)^2,\\ J &= J_0(GH)^2 + c_2L_*AGH\big(AV_0 - 4\xi(4 - 6c_2)Q_*\big)\\ &\qquad - (L_*Q_*)^2 - DF(H^2 - 1). \end{aligned}$$

Then $a_1 = P^2 + J$ again. To prove it, write $J_{DF} = c_2L_*A^2 + J_0$ and $R_0 = 2Ac_2\xi(A^2c_2 + 4 - 6c_2)$, expand, and use the identity for Q_*. This establishes the third centered equation without excluding exceptional values of A, C, ξ.

5.4 Exact degree and verification boundary

For the fixed source, $a = \deg A = 16\,384$, $\deg \Sigma = 81\,922$, $\deg a_0 = 294\,382$, and $\deg a_1 = 458\,818$. The first two centered degrees are $72a + 140$ and $68a + 140$. The deepest center reaches

$$4 \deg \Sigma + \deg a_0 + \deg a_1 = 1\,080\,888.$$

Here the cross-term $-4\Sigma^2a_0T^2$ dominates; further corrections to this same radical alone do not remove that obstruction. The independently authored checker verifies the exact polynomial identities, integrality in halfA, actual numeral payloads, full source-to-equation replays, topology, liveness, and operation counts. Nonzero highest-degree coefficients modulo three primes certify attainment of the stated degrees. Finite replay is an audit of transcription, not a replacement for the universal identity or the inherited consistency calibration. These results are mathematical proofs, not new Lean/Rocq certificates.

6 Independent audit of the centered integer equations

An independently written checker imports no producer builder, circuit helper, evaluator, or checker. It validates all actual fixed numeral payloads and source-port bindings, reconstructs all 22 source target identities across 20 466 kernel interactions, and checks every named literal register by exact polynomial coefficient cancellation. It verifies the universal square completions and norm translation over integer polynomial rings. In particular, the deepest correction is certified in

$\mathbb{Z}[\mathrm{halfA}, C, \xi, B, y]$ after the substitution $A = 2\,\mathrm{halfA}$, not admitted as a divided expression in an unrestricted independent A.

The four independent operation counts are $471, 485, 496, 545$, with respective exact total degrees $1\,245\,324$, $1\,179\,788$, $1\,114\,252$, and $1\,080\,888$. The final exact reuse of $\mathrm{halfU}\,V$ in the halfA and Phalf registers saves one operation without changing their polynomials. Each exact-degree claim has three nonzero leading-coefficient certificates using the actual fixed integers. The checker also performs 448 signed and degenerate modular replays per candidate, including moduli 2 and 3. Normal and optimized Python execution use explicit exceptions, not assertions. These finite checks are regression tests, not substitutes for universal mathematical proofs.

Two logical distinctions matter. First, the square-factor and residue argument recovers positive squares b, Q, and thus positive $\mathcal{C} = bB^{116}$, before the short guard is used to infer $0 < g < \mathcal{C}$ and $4(C - KL)^2 < K^2$. There is no sign assumption hidden in the ten unrestricted integer inputs. Second, replacing the guard preserves existence of solutions through all-signed soundness and canonical-witness completeness; it does not assert a bijection between old and new arbitrary witness tuples. Only the subsequent triangular changes $m \mapsto m + P$ are literal integer bijections, with inverse $m \mapsto m - P$, even when $\Sigma = 0$.

The inherited source/compiler/refutation calibration and Pell–Lucas reliability and interpolation theorems remain explicit ordinary-mathematical dependencies. No new formal-kernel certificate, historical priority, or optimality is claimed. Full receipts, circuit hashes, exact identities, and a proof narrative appear in the `norm_audit` component.

7 Eleven positive variables using two small gaps

The new polynomial has eleven positive variables, degree 211, and 113 live binary arithmetic operations. It improves the incoming centered 257/114 construction in both degree and evaluation cost. The earlier 277/111 point still uses two fewer operations. Centering itself is not new: our contribution is a second center, a cheaper gap bound, and a cofinality argument that permits a useful change of the auxiliary witness.

7.1 The literal polynomial

Use the complete compact44 coefficient tuple β, K_0, M_C, M_F, with M_F already shifted, as bound in `positive/native/ACTIVE_BINDING.json`. The unknowns are

$$j, F, \alpha, i, T, X, Y, n, \eta, \zeta, \gamma \in \mathbb{Z}_{>0}.$$

Set

$$\begin{aligned}
J &= 3j + 1, & q &= \beta J + 1, & C &= q - F - \alpha,\\
E &= XY, & k &= \eta + \zeta, & c &= kY + \eta,\\
a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1,\\
\ell &= EkY, & I &= ic^2, & D_f &= 1 + \Delta I^2,\\
Q &= \Delta(D_f - 1), & \delta &= Q - 1, & D_\tau &= \ell(\ell + k) + 1.
\end{aligned}$$

The coefficient and transport registers are

$$\begin{aligned}
R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,\\
M &= q^3(q-1), \qquad B_t = (qK_0 + X)C - qF.
\end{aligned}$$

Here are the new centers and the small transport gap:

$$T_o = T + q + \alpha, \qquad b = cT_o - R, \quad V = D_f b - c,$$
$$D = (a+2)I, \qquad L = kT_o + F, \quad W = 1 + (q-1)L,$$
$$\overline{Z} = M(\ell + k + n + L + V + D) + (q-1)Y - q^3 W B_t.$$

Keep the adaptive main carrier

$$r_0 = k - R - 1, \qquad S = 4qY, \quad U = qS,$$
$$D_* = U(X + ac) + H(U\gamma + SX - r_0), \qquad A_1 = D_*^2 - U^2(1 + \Delta c^2).$$

Define the three-root norm without any division:

$$A = M^2 D_\tau, \quad B = M^2 D_f, \quad C_N = M^2(QV^2 - 1),$$
$$T_N = \delta^2 AB, \quad R_N = \delta C_N,$$
$$J_N = \delta(\overline{Z}^2 - A - B) - C_N,$$
$$N_c = J_N^2 - 4[T_N + R_N(A + B)],$$
$$N_8 = N_c^2 - 64 T_N R_N \overline{Z}^2,$$
$$\boxed{P_{11} = N_8 + A_1 + N_8 A_1 = 0.}$$

All registers are abbreviations, not additional quantified coordinates.

The exact circuit is `positive/p11_degree211.json`.

7.2 Recover divisibility before using positivity

Write, only in the proof,

$$D_y = \frac{QV^2 - 1}{Q - 1}, \qquad \tau = \sqrt{D_\tau}, \quad f = \sqrt{D_f}, \quad y = \sqrt{D_y}.$$

The inherited unconditional separation lemma applies for every integral b, without assuming $C > 0$ or $V > 0$. Indeed $I \geq c^2$, $c \geq 3$ imply $1 < D_\tau < D_f < Q$ and $|V| > 1$. If $b = 0$, then $V = -c$ and $f > \tau + y$; otherwise $y > f + \tau$. The radicands are distinct, since $D_y = D \in \{D_\tau, D_f\}$ would imply $Q(D - V^2) = D - 1$, with $0 < D - 1 < Q$. Thus no signed sum vanishes.

The norm equation gives $\overline{Z}/M = \tau_* + f_* + y_*$ with the appropriate signed roots. These roots are rational. An explicit extraction identity is as follows: for squares a_1, a_2, a_3 and nonzero sum $s = x_1 + x_2 + x_3$, put

$$u = \frac{s^2 - a_1 - a_2 - a_3}{2}, \qquad v = \frac{u^2 - a_1a_2 - a_1a_3 - a_2a_3}{2s}.$$

Then $x_i = a_i(a_i + u)/(a_i s + v)$; the denominator equals $x_i(x_i + x_j)(x_i + x_k)$ and cannot vanish by distinctness. The first two roots are integral because their squares are integers. The adjacent-Pell rational-integrality lemma makes y_* integral from $QV^2 - (Q-1)y_*^2 = 1$. For completeness, write $y_* = a/m$ reduced; $m^2 \mid Q - 1$, and the norm-preserving descent

$$V' = (2Q-1)V - 2(Q-1)y_*, \qquad y' = (2Q-1)y_* - 2QV$$

has integral V', unchanged reduced denominator m, and $0 < |V'| < |V|$ when $|V| > 1$. The terminal case $|V| = 1$ forces $m = 1$.

Therefore $M \mid \overline{Z}$, and reduction modulo q^3 gives $q^3 \mid Y$. This is the key dependency order: no transport-gap sign or positivity of C has been used yet.

7.3 A degree-two bound for the auxiliary gap

The authentic masks give $0 < (M_C + qM_F)J < 2q^2$ and $q \geq 16$. Directly,

$$-Fq^3 < R < q^3(q+\alpha).$$

For the upper inequality, subtract the bound $q(q-1)(q^2-1)+\alpha(q^2-1)+2q^2$ from $q^3(q+\alpha)$; the difference is $q^3-q^2-q+\alpha(q^3-q^2+1) > 0$. Since $c > Y \geq q^3$, we get

$$c(q+\alpha) > R, \qquad b > cT \geq c, \qquad V > 1.$$

Also $\delta > (\Delta-1)D_f$, $c < 2kY$, and $\Delta - 1 = (a+2)^2 - 2 > 4Y^2$. Consequently

$$\begin{aligned} 0 < y - V = \frac{V^2-1}{\delta(y+V)} &< \frac{b}{\Delta-1} \\ &< \frac{cT_o + Fq^3}{\Delta-1} < kT_o + F = L. \end{aligned}$$

The second center satisfies

$$D^2 - D_f = I^2 - 1 > 0, \qquad 0 < D - f = \frac{I^2-1}{D+f} < I.$$

Thus the huge root y is transported only through $y-V$, and the strong root f through the favorable gap $D-f$.

7.4 Soundness and cofinal completeness

Recover positive witnesses

$$u = y - V, \qquad r = n + L + V + D - f_* - y_*, \qquad f_{\text{old}} = |f_*|, \quad i_{\text{old}} = i.$$

Indeed $r \geq n + L - (y-V) + D - f > 0$. The weighted first carrier with this r and weight L is $M\tau_*$. The strong and auxiliary residuals, and A_1, vanish. The inherited weighted-source theorem therefore yields a finite ZFC refutation. The signed roots, not their absolute values, must be used inside r.

Conversely, canonical source witnesses give the outer coordinates and a normalized strong pair f, I, and the inherited auxiliary lemma supplies one positive pair V_0, y_0 with

$$QV_0^2 - (Q-1)y_0^2 = 1, \qquad V_0 = f^2(cT_0 - R) - c.$$

The matrix

$$B_Q = \begin{pmatrix} 2Q-1 & 2(Q-1) \\ 2Q & 2Q-1 \end{pmatrix}$$

has determinant one, positive entries, and preserves the norm. Some positive power is the identity modulo $h = cf^2$, by finiteness of $\mathrm{SL}_2(\mathbb{Z}/h\mathbb{Z})$. Its iterates give unbounded positive pairs with $V_j \equiv V_0 \pmod{h}$, so

$$T_j = (V_j + c + Rf^2)/(cf^2)$$

is integral and unbounded. Choose $T_o = T_j > q+\alpha$ with $L = kT_o + F > I$. Then $T = T_o - q - \alpha > 0$. The canonical weighted carrier supplies $r > qW - 2k > 2L > L + I$; hence

$$n = r - L - V - D + f + y = r - L - (D-f) + (y-V) > 0.$$

It has $\overline{Z} = M(\tau + f + y)$, and the unchanged adaptive carrier supplies positive γ. This proves completeness without assuming an unjustified size condition on the original auxiliary witness. Finally, N_8 and A_1 are squares modulo four, so their unit-product finalizer vanishes exactly when both vanish. Negating solvability gives Con(ZFC) in ZFC, with the same explicitly inherited compiler and calibration boundary.

7.5 Exact degree rather than a generic gate bound

Put $z = \overline{Z} - M(V + D)$, $e = I^2 - 1$, and $H_0 = z^2 + 2MDz + M^2e - A$. Then

$$J_N = \delta[H_0 + 2MV(z + MD)] - M^2(V^2 - 1).$$

The identity $B = M^2(D^2 - e)$ cancels the apparent degree-102 part of N_c. Its degree-99 head is $8Q_0^2M_0^3V_0^2D_0z_0$, and the norm has degree-198 head

$$64Q_0^4M_0^6V_0^4D_0^2(z_0^2 - M_0^2D_{\tau,0}).$$

With $q_0 = 3\beta j$, $k_0 = \eta + \zeta$, $L_0 = k_0(T + q_0 + \alpha)$ and $B_{t,0} = (K_0q_0 + X)(q_0 - F - \alpha) - q_0F$, we have

$$z_0 = M_0(\ell_0 - L_0B_{t,0}), \qquad z_0^2 - M_0^2D_{\tau,0} = M_0^2L_0B_{t,0}(L_0B_{t,0} - 2\ell_0) \neq 0.$$

Multiplying by the degree-13 adaptive-carrier head shows that the coefficient of

$$j^{48}i^{18}\eta^{39}X^{30}Y^{67}F^4T^2\alpha^2\gamma$$

is $8192(3\beta)^{48} > 0$. This certifies exact degree 211 on the actual coefficient slice, independently of K_0, M_C, M_F.

7.6 A materialized incoming degree 70 companion

The manuscript-only incoming `report.zip` already proposes a 12-positive-variable equation of degree 70 with 100 operations. Its advertised JSON was absent. We supply it as `positive/p12_degree70.json`, with independent checks matching all metrics, but do not claim its theorem as new here. It replaces the normalized strong norm by the ordinary strong norm after an early index bound licenses ordinary rank recovery. Its exact residual degrees are $15, 13, 42$. The full proof is retained verbatim in `positive/interfaces/incoming_report.tex`; the crucial endpoint is the ordinary source, not a possibly nonintegral division of the strong multiplier by Δ.

8 Low-degree integer equations by delayed transport elimination

The new low-degree frontier point established here is

$$(n, d) = (21, 16) \qquad \text{over unrestricted integers.}$$

The completed additional charts $(20, 20), (19, 32), (18, 88)$ improve the initially inspected incoming family, but not the later Library supplement, which already supplies $(20, 20), (19, 28), (18, 32), (17, 88)$. Thus the two $(20, 20)$ charts are alternative derivations, and the $(19, 32), (18, 88)$ charts are dominated comparisons, not new frontier records. They are retained to expose the reversible substitutions. These are variable/degree results, not operation-count or global-minimality claims. Each equation is the sum of squared literal residuals in its packet.

8.1 An exact pivot that becomes inexpensive later

Use the complete source34 tuple, including the already shifted M_F, and the fixed scales $A = 2^{1113}$ and $B = 2^{666104}$. Set

$$q = \beta(3J_3 + 1) + 1, \quad W = 4(3w_3 + 1), \quad C = q - F - \alpha.$$

The parent transport row is

$$(K_0 + W)C - F - z(q - 1) = 0.$$

After replacing α by C, it is monic in F. Hence

$$F = (K_0 + W)C - z(q - 1), \qquad \alpha = q - F - C$$

are integral polynomial restorations; neither $q - 1$ nor any fixed coefficient is divided out. The determinant guard

$$T = AFBC - (q - AF - BC)^2$$

then has degree four, and the packing row degree five. These costs are small compared with the larger degree budgets in the later projection families. This is why delaying the pivot saves a variable without raising their final degree.

8.2 The primary point with 21 integers at degree 16

Start from the exact 24-variable joint-positivity parent. Write $\Delta = a^2+4a+3$ and $S = I^2-\Delta(f^2-1)$. Besides the transport pivot, eliminate $V = fo-c$, shear the auxiliary root $y_{\text{old}} = fo+y$, and replace its old row by

$$A_{\text{new}} = (fo + y)^2 - \Delta(f^2 - 1)(c + y)(2fo + y - c) - 1.$$

The identity

$$A_{\text{new}} = A_{\text{old}} + S(y_{\text{old}}^2 - V^2)$$

is reversible while retaining $S = 0$. Now eliminate $I = (2r+1)c^2$. The three deleted monic definitions leave $24 - 3 = 21$ unknowns and $12 - 3 = 9$ rows. The strong row has degree six, the centered auxiliary row seven, and the joint guard eight. The coefficient certificate is

$$[w_3^8 C^8]P_{21} = 12^8 A^8 > 0.$$

Thus the degree is exactly sixteen on the actual fixed coefficient slice. All integer signs and all parent witnesses are covered.

8.3 The smaller-variable tradeoffs

For these endpoints retain the original auxiliary row instead. Together with the transport pivot and the monic V, I eliminations, apply

n	additional restorations	rows	d
20	$a = E + Y$	8	20
19	$E = XY,\ a = (X + 1)Y$	7	32
18	$E = XY,\ a = (X + 1)Y,\ \eta = c - Yk$	6	88

where $X = qW$, $Y = (2v+1)q^3$, and $k = 1 - V + jc + hE$. These are ordinary polynomial substitutions, not divisibility relaxations. The selected exact leading coefficients are

$$[f^4v^4J_3^{12}]P_{20} = 2^4 3^{12}\beta^{12},$$
$$[\eta^4h^4w_3^4v^4J_3^{16}]P_{19} = 2^{12}3^{20}\beta^{16},$$
$$[h^8w_3^8v^{16}J_3^{56}]P_{18} = 2^{32}3^{64}\beta^{56}.$$

They are strictly positive for the materialized tuple. The eighteen unknowns are exactly

$$J_3, z, f, h, r, v, w_3, \tau, y, \gamma, o, j, c, t, p, q_{\mathrm{pos}}, r_{\mathrm{pos}}, C.$$

A different twenty-variable chart uses one free scale Y_0 with $E = XY_0$, $a = (X+1)Y_0$, retaining $Y_0 - (2v+1)q^3 = 0$. It again has degree twenty, with selected coefficient $36^4\beta^4$ at $\eta^4h^4J_3^4w_3^4Y_0^4$.

8.4 Correctness and checks

Projection and the displayed polynomial restorations are inverse on integer zero sets; the auxiliary shear and row operation are explicitly reversible. Composing them with the pinned parent's mathematical native/Pell compiler theorem yields, in ZFC, nonexistence of an integer zero if and only if Con(ZFC). This is not a new kernel-checked certificate or a proof of consistency.

The package includes all six actual fixed coefficients, complete literal residuals and ordered variable lists, the inherited proof chain, and a separate full proof. A newly authored builder verifies homogeneous-top coefficient certificates. An independent standard-library replay checks all restored rows on 750 signed modular assignments and authenticates all coefficient payloads, in normal and optimized Python. Finite replay is a transcription check, not the unbounded equivalence proof. No downloaded generator is executed.

9 Exact equations and reproducible evidence

The release contains three explicitly defined ordinary-polynomial formats. They differ only in storage, not in logical status.

1. A sparse list entry $[c, [i_1, \ldots, i_s]]$ denotes $cX_{i_1} \cdots X_{i_s}$. Repeated indices repeat factors; the empty list denotes the constant term. Summing the complete list in `direct/fixed_base/equation_integer.json` defines the fully collected quartic, with zero-based indices from 0 to 27,915.
2. A residual packet defines precisely $\sum_j r_j^2$, where every r_j and every intermediate abbreviation is explicitly listed and all exponents are fixed. The packet's ordered variable list is exhaustive. This is used by the 21-integer equation and its alternative charts.
3. A literal arithmetic DAG defines the polynomial at its designated output. Only fixed integer leaves and the declared unknowns enter; every other node is addition, subtraction or multiplication of earlier nodes. All fixed numeral payloads are present. It is a compact exact polynomial, not an existential statement that a suitable polynomial could be generated. This is used by the ten-integer and positive-variable equations.

The direct sparse file itself is included, not merely a generator or a hash. To keep the archive small, redundant intermediate files are regenerated before auditing. Their SHA-256 values are fixed in the replay record. The natural equation is already completely specified by its literal quadratic residual list.

9.1 Verification contract

The validated runtime is Debian 13 Linux x86_64, Python 3.12.14, SymPy 1.14.0 and zlib 1.3.2. Exact compressed-byte regeneration is claimed for this environment, not all Python releases; another gzip implementation may encode the same integer bytes differently. The complete runtime record is `RUNTIME.json`.

The top-level `verify.py` authenticates the manifest and copies the archive into an isolated temporary directory before executing the explicit `REPLAY.json` allowlist. Only newly authored builders, checkers and retained authored packing checks are run. Static upstream Python, Lean, machine data and proof references remain inert. Ordinary and optimized Python runs are both required; the checks use explicit exceptions and do not depend on enabled assertions. Rebuilding must reproduce every literal equation and actual coefficient payload byte for byte.

Independent exact checks cover all Horn branches and constructor wires, every emitted Pell/power/division certificate, the binomial/factorial compositions, the local-source and bounded-CRT bindings, every affine substitution, and the complete collected direct quartic. The other audits check polynomial square completions over integer rings, source coefficient bindings, operation counts, live nodes, exact leading coefficients, signed restorations, and the positive-domain cofinality proof. Mutation negative controls verify that selected damaged circuits, source rows, fixed payloads and degree claims are rejected. The clean extracted ZIP is replayed as a separate packaging test.

A finite modular regression checks a transcription, not a universal theorem. An exact nonzero leading coefficient proves attainment of a degree once its symbolic upper bound has been established. An exact identity check proves that identity but not the imported proof-calculus calibration. These distinct claims are recorded separately in the component audit reports.

9.2 The scope of the consistency claim

For the direct construction, the calibration is the fixed numerical theorem `KEndpoint.c222_iff_zf_refutation` followed by `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. The numerical constructors and rules are retained in the pinned source files. Their mathematical proof uses ordinary syntactic proof transformations and the ZF/ZFC equiconsistency bridge. It does not simulate K215.

The small-variable branches retain their earlier native coefficient/compiler, Pell and machine-based calibration. This report does not pretend that a new coordinate transformation removes those dependencies. Both routes yield ordinary mathematical biconditionals formalizable in ZFC; neither supplies an independently compiled end-to-end object-language ZFC derivation or a new Lean/Rocq build. No implication concerning the truth of consistency is inferred from a search that finds no zero.

10 Further research with concrete stopping points

1. The direct quartic's main cost is generic exponentiation certification. A specialized proof of the fixed-base or factorial subgraphs should be judged by a regenerated ordinary equation, not by the size of an uncompiled source. The current call ledger gives a reproducible starting point.
2. The common witness bound must remain independent of the label bound. Identifying them would make a CRT bound depend circularly on the accumulator it bounds. A successful replacement must prove a noncircular completeness construction for arbitrary finite proofs.

3. Centering only the final large radical reaches a cross-term obstruction at degree 1,080,888 in the present ten-variable coordinates. Further degree reduction requires changing that cross term or another source register, not merely extending the same formal square-root expansion.
4. Combining the Library supplement's CRT transport chart with the centered auxiliary row gives another 21/16 chart, but no better completed metric. Eliminating its retained scale raises the relevant degree; removing its rank register exposes a larger square. An improvement must overcome those explicit degree obstructions while preserving an integer inverse.
5. The 11-positive construction depends on selecting an auxiliary Pell solution in an unbounded congruence orbit. Any sharper gap estimate must preserve that cofinality and the order "integrality, divisibility, then positivity." Assuming a favorable sign before extracting the roots would invalidate soundness.

These are research questions rather than claims of additional completed results. The present deliverables stop at the exact equations, proofs and replayable checks described above.

Sources and attribution

The principal source is Vladimir Reshetnikov's ProveIt repository, with the current comparison pin and per-file authenticated source records in the archive. Earlier local proof packets are retained with their original provenance, licenses and notices. The arithmetic representation sources include the finite Matiyasevich Pell criterion, the explicit power graph, factorial/binomial identities and the bounded-CRT certificate. The inherited Lucas classification, index separation and interpolation interfaces are attributed in the retained proofs to Zhi-Wei Sun. K215/Wade licensing notices apply only to the small-variable source branch and are preserved.

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